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ABSTRACT

We define an endogenous magnetic reconnection process as the one having a driving factor that lies within the region where a drastic change in magnetic field topology occurs. A process of this kind is shown to take place when an electron temperature gradient is present in a plasma column and when, referring to quasi-collisionless regimes, the evolving electron temperature fluctuations are anisotropic. A two-fluid theory, for weakly collisional plasma regimes and confinement configurations which have sheared magnetic fields, is formulated, and two classes of reconnecting modes are identified. The localized class of modes is characterized by a transverse reconnecting magnetic field $\tilde{B}_x$ of odd parity, as a function of the radial variable (across the reconnection layer). The width of this remains significant even when large macroscopic distances, such as those of interest to space and astrophysics, are considered. The relevant reconnection can be referred to as antisymmetric. In fact, the (additional and realistic) presence of a finite current density gradient leads to finding modes with mixed $\tilde{B}_x$ parity which are no longer localized within the reconnection layer. Then, the resulting magnetic islands are not symmetric. In view of the fact that there are plasmas in the Universe with considerable electron thermal energy contents, the features of the novel reconnecting modes can be relied upon in order to produce generation of magnetic fields or conversion of magnetic energy into particle energy (through a sequence of mode-particle resonances).

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I. INTRODUCTION

The main characteristic of an endogenous magnetic reconnection process is that its driving factor lies within the layer where a drastic change in magnetic field topology occurs. This kind of process is shown to take place when an electron temperature gradient is present in a magnetically confined plasma and when, referring to quasi-collisionless regimes, the evolving electron temperature fluctuations are anisotropic.[1] In contrast to this, well known reconnection processes such as those represented by the Sweet-Parker model or the resistive internal kink mode[2] are driven by factors (e.g., flows or plasma pressure gradients) that are outside the reconnection region. A two-fluid theory of the endogenous reconnection that is applicable to weakly collisional plasma regimes is developed for the confinement configuration with significant rates of magnetic field shear. Then,[3] two classes of reconnecting modes are identified. The localized class of modes involves a transverse reconnected field $\tilde{B}_x$ of odd parity (as a function of the radial variable) that is in contrast to the commonly considered reconnecting modes and has characteristic phase velocities and growth rates associated with a finite effective resistivity. The width of the relevant reconnection layer remains significant even when large macroscopic distances are considered. The other class of modes is characterized by $\tilde{B}_x$ of even parity within the reconnection region and extends well beyond it. In fact, the presence of a finite current density gradient leads to finding modes that are composites of odd and even parity $\tilde{B}_x$. In view of the fact that there are plasmas in the Universe with considerable electron thermal energy contents, these features of the considered modes can be relied upon in order to produce generation of magnetic fields or conversion of magnetic energy into particle energy when the coupling of the localized odd modes to the extended even modes can be significant.

An illustrative example of endogenous reconnection is that of "electrostatic" resistive modes whose growth rate is due to the product of the local density gradient and gravitational acceleration. Although in this case, the reconnected field is relatively small, it is necessary to

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identify its profile in order to determine whether the associated modes are acceptable.

We suggest that a sequence of mode-particle resonances[4] starting from the excitation of the oscillatory modes considered here may be proposed[5] for the generation of experimentally observed high energy electron populations following magnetic reconnection events. An attractive feature of processes of this kind is that the considered modes involve relatively large amounts of particle thermal energies and can channel the energy gained, with their excitation, into that of very few particles.

The modes that we have found have significant phase velocities[1,5] with characteristic magnitudes and directions. Therefore, with their excitation, these modes acquire momentum in the direction of the main magnetic field component. Then, the main body of the plasma column is expected to recoil in the opposite direction and the extracted momentum to be transported away from it by the same modes. When these considerations are transferred to toroidal confinement configurations, we may argue that "reconnecting" modes excited within the main body of the plasma can induce a "spontaneous rotation" of it.[6] Relevant experimental observations[7] are consistent with this theoretical indication. A procedure to evaluate the momentum of the modes, which are standing in the x-direction and propagating in the y and z directions, in the considered simplified plane geometry, was outlined already in Ref. 5.

This paper is organized as follows. In Sec. II, we introduce the simplified equilibrium plasma and field configurations that are adopted in our analysis. In Sec. III, we describe the perturbed state that leads to the excitation of modes involving endogenous magnetic reconnection. In Secs. IV and V, we discuss the basic equations that are relevant to the evolution of the perturbed state and are capable of describing the reconnection producing modes of interest. These equations are the electron thermal energy balance equations, the total momentum conservation equation, the electron longitudinal momentum conservation equation, and the electron mass conservation equation that are relevant to the weakly collisional plasma regimes considered here. In the present paper, we simplify these equations by keeping only the most important factors that drive the endogenous reconnection process and provide relevant analytical solutions of them. The expressions for the perturbed transverse and longitudinal (to the equilibrium magnetic field) electron temperatures that are considered to be relevant to quasi-collisionless regimes are given in Sec. IV. In Sec. V, the relevant total momentum conservation equation and the adopted longitudinal electron momentum conservation equation, where the perturbed electron density is determined from a simplified electron mass conservation equation, are presented. In Sec. VI, the basic equation, for modes that are nearly electrostatic, are endogenous (radially localized within the reconnection layer), and describe the onset of antisymmetric reconnection, is given. In Sec. VII, resistive overstable modes are found for the case where a significant effective electrical resistivity could be present and be compatible with the anisotropy of the electron temperature fluctuations. In Sec. VIII, an analysis of the corresponding modes that are found in the case where the ion diamagnetic frequency and the electron "drift" frequency can be neglected is given and the limits of its validity are pointed out. In Sec. IX, two kinds of endogenous modes are found: one with a phase velocity in the direction of the ion diamagnetic velocity and one in the opposite (electron) direction. For the latter kind of mode, the width of the reconnection layer is found to be independent of the inductivity width and to have physically significant values. In Sec. X, we present a simple and illustrative example of endogenous reconnection that involves nearly electrostatic gravitational resistive modes. Since we may associate the tapping of the magnetic field energy by modes for which the physical factor $\propto dJ_z/dx$ is important, we address its effects in Sec. XI after observing that the symmetry of the total momentum conservation equation is broken due to the presence of the term $\propto (dJ_z/dx)$. Consequently, modes that do not have a definite parity and are described by eigenfunctions that are composites of odd and even parity have to be considered. Conclusions are given in Sec. XII.

II. INITIAL UNPERTURBED CONFIGURATION

The simplest equilibrium configuration that we refer is the "sheared field configuration" in a plane geometry represented by $\mathbf{B} \simeq B_0\mathbf{e}_z + B_y(x)\mathbf{e}_y$, where $B_0 \simeq const$ and $B_0^2 \gg B_y^2$. Clearly, the profile of $B_y(x)$ is related to that of the equilibrium current density $J_z(x)$ according to $dB_y/dx = (4\pi/c)J_z(x)$. The electron and ion temperatures, T_e and T_i, are considered to be nearly isotropic in the equilibrium state. The gradients of the particle densities and temperatures are taken to satisfy the equilibrium relation $d(p_e + p_i)/dx + J_zB_y/c = 0$, where $p_e = n_eT_e$ and $p_i = n_iT_i$. We adopt a frame of reference where no $\mathbf{E} \times \mathbf{B}$ velocity exists in the equilibrium state.

The steps needed in order to transfer, with clear limitations, the results obtained for this plane configuration to cylindrical and axisymmetric toroidal configurations have been indicated previously.[3]

III. PERTURBED STATE

The (normal) modes that we consider involve magnetic field perturbations represented by $\hat{\mathbf{B}} = \tilde{\mathbf{B}}(x)\exp(-i\omega t + ik_yy + ik_zz)$, where $k_z \ll k_y$. The gradients of the particle density and temperatures are assumed to be significant for $|x| = |x_0| > 0$, as we choose to analyze modes such that $\mathbf{k}\cdot\mathbf{B}(x = x_0) = k_zB_0 + k_yB_y(x_0) = 0$ and $\mathbf{k}\cdot\mathbf{B} \simeq (k_yB_y')(x - x_0)$ around $x = x_0$, where $B_y' \equiv dB_y/dx$.

We refer to plasmas with pressure $p \ll B^2/8\pi$, i.e., $\beta \ll 1$. Then the magnetic and electric field perturbations are represented by the vector potential $\hat{\mathbf{A}}$ and the scalar potential $\hat{\Phi}$ as

$$\hat{\mathbf{B}} = \nabla\times\hat{\mathbf{A}}, \quad \text{and} \quad \hat{\mathbf{E}} = -\nabla\hat{\Phi} - \frac{1}{c}\frac{\partial}{\partial t}\hat{\mathbf{A}}, \tag{1}$$

where $\hat{\mathbf{A}}$ is primarily in the z-direction, i.e., $\hat{\mathbf{A}} \simeq \hat{A}_z\mathbf{e}_z$. We then have $\tilde{B}_x \simeq ik_y\tilde{A}_z$, $\tilde{B}_y \simeq -d\tilde{A}_z/dx$, $\tilde{E}_x \simeq -d\tilde{\Phi}/dx$, $\tilde{E}_y \simeq -ik_y\tilde{\Phi}$, and $\tilde{E}_z = -ik_z\tilde{\Phi} + i(\omega/c)\tilde{A}_z$. In particular,

$$\tilde{E}_\parallel \simeq -ik_\parallel\tilde{\Phi} + i\frac{\omega}{c}\tilde{A}_z \simeq -ik_\parallel\tilde{\Phi} + i\frac{\omega}{ck_y}\tilde{B}_x, \tag{2}$$

where $k_\parallel \equiv (\mathbf{k}\cdot\mathbf{B})/B = 0$ at $x = x_0$ and $k_\parallel \simeq (k_yB_y'/B)(x - x_0)$ near $x = x_0$.

The components of the $\tilde{\mathbf{E}}\times\mathbf{B}$ drift velocity are $\tilde{u}_{Ex} \simeq -ick_y\tilde{\Phi}/B$, $\tilde{u}_{Ey} \simeq (c/B)(d\tilde{\Phi}/dx)$ and $\tilde{u}_{Ez} = -c(B_y/B^2)(d\tilde{\Phi}/dx)$. We thus find that $\nabla\cdot\hat{\mathbf{u}}_E \simeq 0$, as $(k_z/k_y)(B_y/B_0) \ll 1$. Introducing $\tilde{\xi}_x \equiv \tilde{u}_{Ex}/(-i\omega)$, we can then rewrite $\tilde{E}_\parallel$ as follows:

$$\tilde{E}_\parallel \simeq \frac{\omega}{ck_y}\left[\tilde{B}_x - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x\right]. \tag{3}$$

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The evolution of the electron temperature T_e in the perturbed state is considered to be anisotropic, i.e., $\hat{T}_{e\perp} \neq \hat{T}_{e\parallel}$, on account of the large anisotropy of the electron thermal conductivity. Then, we note that

$$\nabla \cdot \mathbb{P}_e = \nabla p_{e\perp} + \left[\mathbf{B} \cdot \nabla\left(\bar{\bar{p}}_e \mathbf{B}\right)\right]/4\pi, \tag{4}$$

where $\mathbb{P}_e$ is the electron pressure tensor and

$$\bar{\bar{p}}_e = \frac{4\pi(p_{e\parallel} - p_{e\perp})}{B^2}. \tag{5}$$

Consequently

$$\nabla \cdot \hat{\mathbb{P}}_e \simeq \nabla \hat{p}_{e\perp} + n\mathbf{B}\left[\mathbf{B} \cdot \nabla\left(\hat{T}_{e\parallel} - \hat{T}_{e\perp}\right)\right]/B^2. \tag{6}$$

The ion temperature T_i, on the other hand, is taken to be nearly isotropic in the equilibrium state as well as in the perturbed state. That is, we take

$$\nabla \cdot \hat{\mathbb{P}}_i \simeq \nabla \hat{p}_i. \tag{7}$$

Then, we proceed with equations derived from the total momentum conservation equation, the electron longitudinal momentum conservation equation, the electron mass conservation equation, and the electron thermal energy balance equations that are relevant to the evolution of the perturbed state.

IV. ELECTRON THERMAL ENERGY BALANCE EQUATIONS: SIMPLEST LIMITS

We refer to weakly collisional regimes where ν_{ee}, the electron-electron collision frequency, is considerably smaller than $|\omega|$. Since $|\omega|^2 \ll \Omega_{ce}^2$, where Ω_{ce} is the electron cyclotron frequency, we are justified in using the "equation of state"

$$\frac{d}{dt}\left(\frac{p_{e\perp}}{nB}\right) = 0, \tag{8}$$

where d/dt is the total time derivative. The perturbed form of Eq. (8) yields

$$-i\omega\tilde{T}_{e\perp} + \tilde{u}_{Ex}T_e' \simeq 0, \tag{9}$$

for $|\tilde{B}_z/B| \ll |\tilde{T}_{e\perp}/T_e|$, where $T_e' \equiv dT_e/dx$. That is,

$$\tilde{T}_{e\perp} \simeq -\tilde{\xi}_x T_e', \tag{10}$$

for $-i\omega\tilde{\xi}_x = \tilde{u}_{Ex}$.

Under the condition $\nu_{ee} \ll |\omega|$, $\hat{T}_{e\parallel}$ is decoupled from $\hat{T}_{e\perp}$ and the relevant equation for $\hat{T}_{e\parallel}$, considering the effective large thermal conductivity limit, is

$$0 \simeq D^e_{\parallel\parallel}\nabla_\parallel\left(\nabla_\parallel \hat{T}_{e\parallel} + \frac{\hat{B}_x}{B}T_e'\right) + D^e_{\perp\parallel}\nabla_\perp^2 \hat{T}_{e\parallel}. \tag{11}$$

It corresponds to

$$D^e_{\perp\parallel}\left(\frac{d^2}{dx^2} - k_y^2\right)\tilde{T}_{e\parallel} - k_\parallel^2 D^e_{\parallel\parallel}\tilde{T}_{e\parallel} \simeq -ik_\parallel D^e_{\parallel\parallel}\frac{\tilde{B}_x}{B}T_e'. \tag{12}$$

Here, $D^e_{\parallel\parallel}$ is the longitudinal electron thermal diffusion coefficient, $D^e_{\perp\parallel}$ is the transverse electron thermal diffusion coefficient that could result from the excitation of relatively small scale electrostatic fluctuations, and the condition for the large thermal conductivity limit is $k_\parallel^2 D^e_{\parallel\parallel} \gg |\omega|$. Clearly, Eq. (12) indicates that when $\tilde{B}_x(x = x_0) \neq 0$, $\tilde{T}_{e\parallel}$ would be singular without the transverse diffusion term, if $T_e' \neq 0$. In fact, this transverse diffusion term is retained in the analysis in order to deal with the singularity when we discuss in Sec. XI the "extended" modes that are characterized by $\tilde{B}_x(x)$ with even parity in $x - x_0$ and $\tilde{B}_x(x = x_0) \simeq const$. On the other hand, the localized modes that we identify in Secs. VI–IX are characterized by $\tilde{B}_x(x)$ that are odd in $x - x_0$. For such modes, there is no singularity in $\tilde{T}_{e\parallel}$ and, for their description, we omit the transverse diffusion term for simplicity of presentation. That is, we consider

$$\tilde{T}_{e\parallel} \simeq \frac{i}{k_\parallel}\frac{\tilde{B}_x}{B}T_e', \tag{13}$$

which follows from Eq. (12).

V. MOMENTUM CONSERVATION EQUATIONS

The perturbed total momentum conservation equation can be written as

$$-i\omega n m_i \hat{\mathbf{u}}_i = -\nabla\left(\hat{p}_i + \hat{p}_{e\perp} + \frac{\hat{\mathbf{B}}\cdot\mathbf{B}}{4\pi}\right) - \frac{n}{B^2}\mathbf{B}\left[i(\mathbf{k}\cdot\mathbf{B})\left(\hat{T}_{e\parallel} - \hat{T}_{e\perp}\right)\right] + \frac{1}{4\pi}\left[i(\mathbf{k}\cdot\mathbf{B})\hat{\mathbf{B}} + \hat{B}_x\frac{d}{dx}\mathbf{B}\right]. \tag{14}$$

Applying the operator $\mathbf{B}\cdot\nabla\times$ to Eq. (14) and considering modes for which $\nabla_\perp \cdot \hat{\mathbf{u}}_i \simeq 0$, we find that the equation of interest within $|x - x_0| \sim \delta_m$, where δ_m is the width of the reconstruction layer, is

$$-\omega(\omega - \omega_{di})\left(\frac{d^2\tilde{\xi}_x}{dx^2} - k_y^2\tilde{\xi}_x\right) + \frac{1}{m_i n}\frac{k_y B_y}{B^2}\frac{d}{dx}\left[n(\mathbf{k}\cdot\mathbf{B})\left(\tilde{T}_{e\parallel} - \tilde{T}_{e\perp}\right)\right] \simeq \frac{i}{4\pi m_i n}\left[(\mathbf{k}\cdot\mathbf{B})\left(\frac{d^2\tilde{B}_x}{dx^2} - k_y^2\tilde{B}_x\right) - k_y\frac{d^2B_y}{dx^2}\tilde{B}_x\right]. \tag{15}$$

Here, we have taken $\tilde{u}_{ix} = -i(\omega - \omega_{di})\tilde{\xi}_x$ with $\omega_{di} \equiv k_y[c/(enB)](dp_i/dx)$. We note that $d^2B_y/dx^2 = (4\pi/c)(dJ_z/dx)$, J_z being the equilibrium current density.

The adopted longitudinal electron momentum conservation equation that has a key role in the onset of magnetic reconnection is

$$-i\omega m_e n\hat{u}_{e\parallel} = -\nabla_\parallel \hat{p}_{e\parallel} - \left(\frac{dp_e}{dx}\right)\frac{\hat{B}_x}{B} - en\left[\hat{E}_\parallel - (\eta_\parallel - i\omega\mathcal{L}_I)\hat{J}_\parallel\right], \tag{16}$$

where $\hat{p}_{e\parallel} = \hat{n}_e T_e + n\hat{T}_{e\parallel}$, and $\eta_\parallel$ is a small but finite electrical resistivity. The inductivity $\mathcal{L}_I$ is introduced[8] to take into account the influence, on the plasma current carrying channels associated with magnetic reconnection, of current carrying channels not included in the adopted plasma electromagnetic equation. A finite electron inertia inductivity is represented by the l.h.s. of Eq. (16) as $\tilde{J}_\parallel \simeq -en\tilde{u}_{e\parallel}$ and, in this case, the corresponding inductivity is

$$\mathcal{L}_e = \frac{4\pi}{\omega_{pe}^2} \equiv \frac{4\pi}{c^2}d_e^2. \tag{17}$$

The simplified electron mass conservation equation

$$-i\omega\tilde{n}_e + \tilde{u}_{Ex}\frac{dn}{dx} \simeq 0 \tag{18}$$

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gives $\tilde{n}_e = -\tilde{\xi}_x dn/dx$ as $\tilde{u}_{Ex} = -i\omega\tilde{\xi}_x$.

Next, we note that

$$\tilde{J}_\parallel \simeq \frac{ic}{4\pi k_y}\left(\frac{d^2}{dx^2} - k_y^2\right)\tilde{B}_x, \tag{19}$$

when we consider $(k_z/k_y)(B_y/B_0) \ll 1$. Then, using Eqs. (3), (18), and (19), Eq. (16) becomes

$$(\omega - \omega_{*e} - \omega_{*T})\tilde{B}_x \simeq i(\mathbf{k}\cdot\mathbf{B})\left[(\omega - \omega_{*e})\tilde{\xi}_x + \omega_{*T}\frac{\tilde{T}_{e\parallel}}{T'_e}\right] + \omega\left[d_I^2 + \left(d_e^2 + i\frac{D_m}{\omega}\right)\right]\left(\frac{d^2}{dx^2} - k_y^2\right)\tilde{B}_x, \tag{20}$$

where d_I is the "inductive skin depth" defined by

$$\mathcal{L}_I \equiv \frac{4\pi}{c^2}d_I^2, \tag{21}$$

and $D_m \equiv c^2\eta_\parallel/(4\pi)$ is the magnetic diffusion coefficient that in the collisional limit is $\alpha_\eta \nu_{ei} d_e^2$ where α_η is an appropriate numerical coefficient. The two characteristic frequencies appearing in Eq. (20) are

$$\omega_{*e} \equiv -k_y\frac{cT_e}{eB}\frac{n'}{n} \quad \text{and} \quad \omega_{*T} \equiv -k_y\frac{cT'_e}{eB}, \tag{22}$$

where $n' \equiv dn/dx$. The term d_e^2 arises from the electron inertial term in Eq. (16). We note that, for ultrarelativistic electrons found in astrophysical plasmas this term can be significant.

Using Eq. (13) for $\tilde{T}_{e\parallel}$, Eq. (10) for $\tilde{T}_{e\perp}$, and introducing $\nabla_\perp^2 \equiv d^2/dx^2 - k_y^2$, $\tilde{Z}_x \equiv \tilde{B}_x/B'_y$ and $\tilde{Y}_x \equiv \tilde{\xi}_x k_y$, we rewrite Eq. (20) as

$$\tilde{Z}_x - i(x - x_0)\tilde{Y}_x \simeq \frac{\omega\left(d_I^2 + d_e^2\right) + iD_m}{\omega - \omega_{*e}}\nabla_\perp^2\tilde{Z}_x, \tag{23}$$

while Eq. (15) becomes

$$-\omega(\omega - \omega_{di})\frac{1}{k_y^2}\nabla_\perp^2\tilde{Y}_x - i\omega_{sA}^2\frac{d}{dx}\left[\tilde{Z}_x - i(x - x_0)\tilde{Y}_x\right] \simeq i\omega_H^2\left[(x - x_0)\left(\nabla_\perp^2\tilde{Z}_x\right) - \left(\frac{1}{B'_y}B''_y\right)\tilde{Z}_x\right], \tag{24}$$

where $B''_y \equiv d^2B_y/dx^2$,

$$\omega_H^2 \equiv \frac{\left(B'_y\right)^2}{4\pi n m_i} \quad \text{and} \quad \omega_{sA}^2 \equiv -\frac{T'_e}{2m_i}\frac{d}{dx}\left(\frac{B_y}{B}\right)^2. \tag{25}$$

In particular, $\omega_{sA}^2 = (T_e/m_i)(B_y/B)^2/(r_{T_e}r_{B_y})$ where $1/r_{T_e} \equiv -T'_e/T_e$ and $(1/r_{B_y}) \equiv B'_y/B_y$, and the ratio

$$\varepsilon_A \equiv \frac{\omega_{sA}^2}{\omega_H^2} = \frac{4\pi n T_e}{B^2}\frac{r_{B_y}}{r_{T_e}} \tag{26}$$

is considerably smaller than unity in well confined plasmas.

VI. ANTISYMMETRIC RECONNECTION BY ENDOGENOUS LOCALIZED MODES

We note that localized modes are characterized by

$$\left|\frac{d^2\tilde{B}_x}{dx^2}\frac{1}{\tilde{B}_x}\right| \sim \frac{1}{\delta_m^2}, \tag{27}$$

where δ_m represents the width of the layer where reconnection takes place. A subcategory of this class of modes are the "quasi-electrostatic" ones for which $\nabla \times \hat{\mathbf{E}} \simeq 0$.

In the case under consideration modes of this kind that are illustrated in Sec. X involve $\tilde{Y}_x$ that are antisymmetric functions of $(x - x_0)$ and are found for $|\varepsilon_A| \ll 1$. They are represented by

$$\tilde{Y}_x \simeq \frac{i}{x - x_0}\frac{\omega\left(d_I^2 + d_e^2\right) + iD_m}{\omega - \omega_{*e}}\frac{d^2\tilde{Z}_x}{dx^2}, \tag{28}$$

implying that

$$\delta_m^2 \ll \left|\frac{\omega\left(d_I^2 + d_e^2\right) + iD_m}{\omega - \omega_{*e}}\right|. \tag{29}$$

Moreover, referring to Eq. (24), $\delta_m \ll |B'_y/B''_y|$ and it reduces to

$$-\omega(\omega - \omega_{di})\frac{1}{k_y^2}\nabla_\perp^2\tilde{Y}_x \simeq (x - x_0)^2\frac{(\omega - \omega_{*e})\omega_H^2}{\left(d_I^2 + d_e^2\right)\omega + iD_m}\tilde{Y}_x + \omega_{sA}^2\frac{d}{dx}\left[(x - x_0)\tilde{Y}_x\right]. \tag{30}$$

We observe that the validity of Eq. (28) implies that

$$|\hat{T}_{e\parallel}| \ll |\hat{T}_{e\perp}|. \tag{31}$$

VII. "RESISTIVE" MODES

The simplest case to analyze is that for which $|\omega| > |\omega_{di}| \sim |\omega_{*e}|$, D_m is considerably larger than that given by the collisional theory and we can consider $D_m/|\omega(d_I^2 + d_e^2)| \gg 1$. Then, for the sake of simplicity we take $k_y^2\delta_m^2 \ll 1$ and if we define

$$\bar{x} = \frac{x - x_0}{\delta_m} \text{ and } \Gamma^2 = -\left(\frac{\omega}{k_y\delta_m\omega_{sA}}\right)^2.$$

Equation (30) becomes

$$\Gamma^2\frac{d^2}{d\bar{x}^2}\tilde{Y}_x - \Gamma\bar{x}^2\tilde{Y}_x \simeq \frac{d}{d\bar{x}}\left(\bar{x}\tilde{Y}_x\right), \tag{32}$$

when we take

$$\delta_m \equiv \left|\frac{D_m\omega_{sA}}{\omega_H^2 k_y}\right|^{\frac{1}{3}} = \left|\frac{D_m}{k_y\omega_H}\hat{\varepsilon}_A^{\frac{1}{3}}\right|. \tag{33}$$

Then, it is easy to see that Γ cannot be real. In particular, the solution

$$\tilde{Y}_x = \tilde{Y}_x^0(1 - \lambda\bar{x}^2)\exp\left(-\frac{\sigma}{2}\bar{x}^2\right) \tag{34}$$

with $\mathrm{Re}\sigma > 0$, leads to find

$$\Gamma^2(\sigma + 2\lambda) = -1,$$
$$\Gamma^2\sigma(\sigma + 5\lambda) + (\sigma + 3\lambda) = \Gamma,$$
$$\Gamma^2\sigma^2 + \sigma = \Gamma.$$

Therefore, we obtain $\lambda = \sigma/3$, implying that

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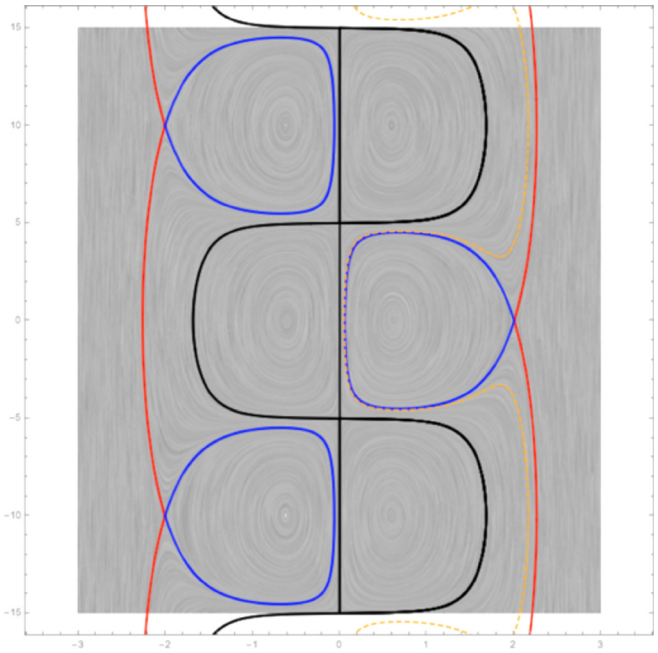


FIG. 1. Strings of magnetic islands produced by the $\tilde{B}_x$-odd localized mode identified in the present paper. Here, x stands for the radial distance from the considered magnetic surface where $\mathbf{k}\cdot\mathbf{B}(x)=0$. The illustrated magnetic field lines involve the radial and vertical components of the (total) reconnected field.

$$\tilde{Z}_x \simeq \tilde{Z}_x^0 \bar{x}\exp\left(-\frac{\sigma}{2}\bar{x}^2\right). \tag{35}$$

Moreover, $\Gamma^3=-6/25$ and $\sigma=-3/(5\Gamma^2)=5/(2\Gamma)$. Consequently, the mode growth rate and frequencies are given by $\Gamma=(6/25)^{\frac{1}{3}}\left(1\pm i\sqrt{3}\right)/2$, and

$$\omega \simeq i\Gamma\left|D_m k_y^2\frac{\omega_{sA}^4}{\omega_H^2}\right|^{\frac{1}{3}}. \tag{36}$$

Clearly, these are overstable modes with larger frequencies than their growth rates as $\mathrm{Re}\omega=\sqrt{3}\mathrm{Im}\omega$. An important feature of the expression for the growth rate is that it shows that the electron temperature gradient is the driving factor of the instability and is proportional to $|dT_e/dx|^{\frac{2}{3}}$. Now, we refer to the magnetic configuration characterized by a transverse field of the kind given by Eq. (35) as antisymmetric reconnection. The strings of relevant magnetic islands are represented in Fig. 1.

VIII. "INDUCTIVE" AND "INERTIAL" MODES

When a process involving a magnetic field diffusion represented by a coefficient D_m cannot be assumed to be present, it is worth considering the purely oscillatory modes that are found when electron inertia and/or an inductivity can be the main physical factor to break the "frozen-in-law."

Then, referring to the analysis presented in Secs. VI and VII, and adopting the simplifying assumptions considered there, we are led to solve the following differential equation for $\tilde{Y}_*\equiv k_y\tilde{\xi}_x$ in the reconnection region

$$\frac{\omega^2}{\omega_H^2 k_y^2\left(\delta_m^0\right)^2}\frac{d^2}{d\bar{x}^2}\tilde{Y}_* + \varepsilon_A\frac{d}{d\bar{x}}\left(\bar{x}\tilde{Y}_*\right) \simeq -\bar{x}^2\tilde{Y}_*\frac{\left(\delta_m^0\right)^2}{\left(d_I^2+d_e^2\right)}, \tag{37}$$

where $\bar{x}\equiv(x-x_0)/\delta_m^0$. In addition, we define

$$\left(\delta_m^0\right)^2 \equiv \varepsilon_A\left(d_I^2+d_e^2\right). \tag{38}$$

Guided by the analysis presented in Sec. VII, we consider

$$\tilde{Y}_*(\bar{x}) = \tilde{Y}_*^0(1-\lambda\bar{x}^2)\exp\left(-\frac{1}{2}\sigma\bar{x}^2\right), \tag{39}$$

as the lowest acceptable eigensolution. Substituting this solution into Eq. (37) and equating the coefficients of the powers of $\bar{x}$, we find $\lambda=\sigma/3$ and $\sigma=5/2$, and the relevant frequency is given by

$$\omega^2 = \frac{6}{25}\frac{\omega_{sA}^4}{\omega_H^2}k_y^2\left(d_I^2+d_e^2\right)\equiv\omega_0^2. \tag{40}$$

When the contribution of D_m is included in the limit where $D_m/|\omega|\ll\left(d_I^2+d_e^2\right)$ we find that $\omega=\omega_0+i\gamma$ with

$$\gamma \simeq \frac{1}{2}\frac{D_m}{\left(d_I^2+d_e^2\right)}, \tag{41}$$

that is a resistive growth rate. We note that the excited modes can have two opposite phase velocities with the same growth rate γ like the purely resistive modes treated in Sec. VII.

As can be expected, when the contribution of the electron-electron collision frequency ν_{ee} is included, a damping factor has to be added to Eq. (40). Therefore, if a purely collisional D_m is considered, a growth rate may not be found.

The present analysis has assumed that both the electron drift frequency ω_{*e} and the ion diamagnetic frequency ω_{di} are negligible when compared to ω_0. The relevant condition $\omega_0^2\gg\omega_{di}^2$ implies that

$$\left(\frac{6}{25}\right)\left(\frac{1}{r_{T_e}}\frac{T_e}{m_i}\right)^2\left[\frac{1}{2}\frac{d}{dx}\left(\frac{B_y}{B}\right)^2\right]^2\frac{4\pi n m_i}{\left(B_y'\right)^2}k_y^2\left(d_I^2+d_e^2\right) > k_y^2T_i^2\frac{1}{r_{pi}^2}\left(\frac{c}{eB}\right)^2, \tag{42}$$

where $1/r_{pi}=-d\ln p_i/dx$. This means that

$$\left(d_I^2+d_e^2\right)\left\{\left(\frac{6}{25}\right)\left(\frac{T_e}{T_i}\right)^2\frac{r_{pi}^2}{r_{T_e}^2}\frac{B_y^2}{B^2}\right\} > \left(\frac{c}{\omega_{pi}}\right)^2 \tag{43}$$

and it can be a rather severe condition on d_I and/or d_e.

In the case of the resistive modes treated in Sec. VII, referring to Eq. (36), the necessary condition to neglect $\omega_{di}\sim\omega_{*e}$ is

$$\left|\frac{\omega_{sA}^4}{\omega_H^2}D_m k_y^2\right| > k_y^3\mathrm{v}_{di}^3 \tag{44}$$

and this means

$$D_m > |k_y\mathrm{v}_{di}|\frac{\mathrm{v}_{di}^2\omega_H^2}{\omega_{sA}^4} = |k_y\mathrm{v}_{di}|\left(\frac{c^2}{\omega_{pi}^2}\right)\left[\frac{T_i}{T_e}\frac{B}{B_y}\frac{r_{T_e}}{r_{pi}}\right]^2, \tag{45}$$

where $\mathrm{v}_{di}\equiv cT_i/(eBr_{pi})$.

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IX. LOCALIZED MODES INVOLVING DIAMAGNETIC AND DRIFT FREQUENCIES

Now, we refer to the case where the frequencies ω_{di} and ω_{*e} cannot be neglected in comparison with ω while, for the sake of simplicity, we take at first $D_m/|\omega| \ll (d_I^2 + d_e^2)$ and consider the long wave limit $k_y^2\delta_m^2 \ll 1$. Then, Eq. (30) reduces to

$$-\omega(\omega-\omega_{di})\frac{1}{k_y^2}\frac{d^2\tilde{Y}_x}{dx^2} \simeq (x-x_0)^2\frac{(\omega-\omega_{*e})\omega_H^2}{(d_e^2+d_I^2)\omega}\tilde{Y}_x + \omega_{sA}^2\frac{d}{dx}\left[(x-x_0)\tilde{Y}_x\right]. \tag{46}$$

The solution of this equation that corresponds to the acceptable profile

$$\tilde{B}_x \simeq \tilde{B}_x^0(x-x_0)\exp\left[-\frac{\sigma}{2}(x-x_0)^2\right], \tag{47}$$

is of the form $\tilde{Y}_x \simeq \tilde{Y}_x^0[1-\lambda(x-x_0)^2]\exp[-\sigma(x-x_0)^2/2]$ where $\lambda = \sigma/3$ and $\sigma = (3/5)(k_y^2\omega_{sA}^2)/[\omega(\omega-\omega_{di})]$. Moreover, the solution requires that

$$(\omega-\omega_{*e})(\omega-\omega_{di})\omega_H^2 = \frac{6}{25}k_y^2(d_e^2+d_I^2)\omega_{sA}^4. \tag{48}$$

We consider the mode with a phase velocity in the direction of the electron diamagnetic velocity to be of particular relevance as in this case the width δ_m of the layer over which the mode is localized is independent of $d_e^2 + d_I^2$ and remains significant even when the macroscopic distances entering the theory are large. Therefore, this mode remains significant even when only the finite electron inertia inductivity is taken into account. In particular, we obtain

$$\omega \simeq \omega_{*e} + \delta\omega, \qquad \delta\omega \simeq \frac{6}{25}(d_e^2+d_I^2)\frac{\omega_{sA}^2}{v_{*e}(v_{*e}-v_{di})}\omega_{*e}, \tag{49}$$

$$\delta_m^2 = \frac{1}{\sigma} = \frac{5}{3}\frac{v_{*e}(v_{*e}-v_{di})}{\omega_{sA}^2} = \frac{5}{3}\left(\frac{cm_i}{eB_y}\right)^2\frac{T_e}{m_i}\left(1+\frac{T_i}{T_e}\frac{r_n}{r_{pi}}\right)\frac{r_{T_e}r_{B_y}}{r_n^2}, \tag{50}$$

where $v_{*e} = \omega_{*e}/k_y$ and $1/r_n \equiv -(n'/n)$. We observe that δ_m is of the same order as the ion gyroradius in the field B_y that is considerably larger than the gyroradius (in the total field). We can verify that as long as $\varepsilon \ll 1$ the adopted quasi-electrostatic approximation is valid.

If we add a finite magnetic diffusion coefficient to the relevant analysis, we find that a growth rate is associated with this. In particular we obtain, instead of Eq. (49)

$$\delta\omega \simeq C_D\left(d_e^2+d_I^2+i\frac{D_m}{\omega_{*e}}\right)\omega_{*e}, \tag{51}$$

where C_D is given by Eq. (49) and the relevant growth rate is $\gamma \simeq C_D D_m$.

Referring to the mode with $\omega \simeq \omega_{di} + \delta\omega$ Eq. (46) may be rewritten as

$$-\omega_{di}\delta\omega\frac{1}{k^2}\frac{d^2\tilde{Y}_x}{dx^2} \simeq (x-x_0)^2\frac{(\omega_{di}-\omega_{*e})\omega_H^2}{\omega_{di}(d_e^2+d_I^2)}\tilde{Y}_x + \omega_{sA}^2\frac{d}{dx}\left[(x-x_0)\tilde{Y}_x\right]. \tag{52}$$

Then, we can see that

$$\delta_m^2 \sim (d_e^2+d_I^2)\frac{\omega_{sA}^2}{\omega_H^2}, \tag{53}$$

for $T_e \sim T_i$, that can be very small. Moreover

$$\left|\frac{\delta\omega}{\omega_{di}}\right| \sim (d_e^2+d_I^2)\frac{\omega_{sA}^4}{v_{di}^2\omega_H^2}. \tag{54}$$

X. ILLUSTRATIVE CASE OF ANTISYMMETRIC AND ENDOGENOUS RECONNECTION

The simplest example of Antisymmetric and Endogenous Reconnection that can be considered is that of the quasi-electrostatic gravitational mode[9] that can be found in the presence of a relatively large D_m. The analysis starts from the same stationary configuration as the one that is introduced in Sec. II to which a gravitational force represented by the acceleration $\mathbf{g} = -g\mathbf{e}_x$ is added. For simplicity, we neglect the effects of ω_{di} and ω_{*e}. Then, applying the operator $\mathbf{e}_z\cdot\nabla\times$ to the perturbed total momentum conservation equation, where we take $\hat{n} \simeq -(dn/dx)\hat{\xi}_x$, we find

$$\gamma^2\frac{d^2\tilde{\xi}_x}{dx^2} \simeq \frac{i}{4\pi m_i n}(\mathbf{k}\cdot\mathbf{B})\frac{d^2\tilde{B}_x}{dx^2} - k_y^2\frac{g}{r_n}\tilde{\xi}_x \tag{55}$$

for $k_y^2(\delta_m^g)^2 \ll 1$, and $\delta_m^g \ll B_y'/B_y''$, δ_m^g being the relevant characteristic distance. Here, $\gamma \equiv -i\omega$.

For the perturbed longitudinal electron momentum conservation equation, we take simply $0 \simeq -en(\hat{E}_\parallel - \eta_\parallel\hat{J}_\parallel)$. Then, referring to Eqs. (3) and (9) we have

$$0 \simeq i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x + \frac{D_m}{\gamma}\frac{d^2\tilde{B}_x}{dx^2}, \tag{56}$$

an approximation that is valid for $\gamma_g^2 \equiv g/r_n \ll \omega_H^2$. We are also assuming that $k_y^2 \ll d^2/dx^2$. Next, considering $\mathbf{k}\cdot\mathbf{B} \simeq k_y B_y'(x-x_0)$, introducing $\bar{\bar{x}} \equiv (x-x_0)/\delta_m^g$ and defining $\tilde{\bar{B}}_x \equiv i\tilde{B}_x/(k_y\delta_m^g B_y')$ we obtain the eigenvalue equation

$$\frac{d^2\tilde{\xi}_x}{d\bar{\bar{x}}^2} + (\lambda - \bar{\bar{x}}^2)\tilde{\xi}_x \simeq 0, \tag{57}$$

where

$$\lambda \equiv \frac{k_y^2(\delta_m^g)^2}{\gamma^2}\frac{g}{r_n}, \quad (\delta_m^g)^4 = \left(\frac{D_m\gamma}{k_y^2\omega_H^2}\right), \tag{58}$$

and

$$\frac{d^2\tilde{\bar{B}}_x}{d\bar{\bar{x}}^2} \simeq \frac{\gamma(\delta_m^g)^2}{D_m}\bar{\bar{x}}\tilde{\xi}_x. \tag{59}$$

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Clearly, Eq. (57) has a spectrum of eigensolutions of the form $H_n(x)\exp(-\bar{\bar{x}}^2/2)$, where H_n's are the Hermite polynomials of order n and the eigenvalues are given by $\lambda = 2n+1$. For $n=0$, $\tilde{\bar{\xi}}_x = \exp(-\bar{\bar{x}}^2/2)$ and $\lambda = 1$ implying $\gamma > 0$. However, Eq. (59) indicates that $d^2\tilde{\bar{B}}_x/d\bar{\bar{x}}^2$ and, hence, the curvature of $\tilde{\bar{B}}_x$ is positive for all $\bar{\bar{x}} > 0$, if we consider the fact that $\tilde{\bar{B}}_x$ is odd in $\bar{\bar{x}}$. This is not acceptable for a bound solution to exist. Moreover, the $n=1$ eigensolution, $\tilde{\bar{\xi}}_x = \bar{\bar{x}}\exp(-\bar{\bar{x}}^2/2)$ and $\lambda = 3$, implying $\gamma > 0$, is not consistent with a localized $\tilde{B}_x$ function if $g/(r_n\omega_H^2)$ is small. In this case, $\tilde{B}_x$ extends beyond the reconnection region. Then, the lowest acceptable localized eigensolution corresponds to $n=2$ and, in this case

$$\tilde{\bar{\xi}}_x(\bar{\bar{x}}) = \tilde{\bar{\xi}}_x^0(1-2\bar{\bar{x}}^2)\exp\left(-\frac{\bar{\bar{x}}^2}{2}\right). \tag{60}$$

The growth rate of the relevant mode, corresponding to $\lambda = 5$, is given by

$$\gamma^3 = \frac{1}{25}\left(\frac{g}{r_n}\right)^2\frac{D_m k_y^2}{\omega_H^2}. \tag{61}$$

An approximate solution that reproduces the asymptotic limits of $\tilde{B}_x$ for $\bar{\bar{x}}^2 \ll 1$ and $\bar{\bar{x}}^2 \gg 1$ is

$$\tilde{\bar{B}}_x \simeq -\tilde{\bar{\xi}}_x^0\bar{\bar{x}}\frac{1+2\bar{\bar{x}}^4}{1-\bar{\bar{x}}^2/3+\bar{\bar{x}}^4}\exp\left(-\frac{\bar{\bar{x}}^2}{2}\right). \tag{62}$$

We note that without an analysis of the profile of $\tilde{\bar{B}}_x(\bar{\bar{x}})$, although the mode is nearly electrostatic, we could not have identified the lowest physically acceptable eigensolution.

XI. COMPOSITE SOLUTION AND ASSOCIATED ASYMMETRIC RECONNECTION

The presence of the current density gradient term, $d^2B_y/dx^2 \propto dJ_z/dx$, in Eq. (15) introduces an even parity component for $\tilde{B}_x(x-x_0)$ to be added to the "endogenous" solution given, for instance, by Eq. (47). We represent this odd component as

$$\tilde{B}_x^{od} = \tilde{B}_0^{od}\bar{x}W_0(\bar{x}^2), \tag{63}$$

and add the component

$$\tilde{B}_x^{ev} = \tilde{B}_0^{ev}\left[1+\varepsilon_*\varphi_0(\bar{x}^2)\right], \tag{64}$$

where $\bar{x} \equiv (x-x_0)/\delta_m$, and $\varepsilon_* \ll 1$.

In order to identify a consistent composite solution, we start from the even component of Eq. (15), that is,

$$-\omega(\omega-\omega_{di})\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{\xi}_x^{ev} + \frac{1}{m_i}\frac{k_yB_y}{B^2}\frac{d}{dx}\left[(\mathbf{k}\cdot\mathbf{B})\left(\tilde{T}_{e\|}^{ev}-\tilde{T}_{e\perp}^{ev}\right)\right] \simeq \frac{i}{4\pi m_i n}\left[(\mathbf{k}\cdot\mathbf{B})\left(\frac{d^2\tilde{B}_x^{od}}{dx^2}-k_y^2\tilde{B}_x^{od}\right) - k_y\frac{d^2B_y}{dx^2}\tilde{B}_x^{ev}\right], \tag{65}$$

where

$$(\mathbf{k}\cdot\mathbf{B})\left(\tilde{T}_{e\|}^{ev}-\tilde{T}_{e\perp}^{ev}\right) \simeq iT_e'\left[\tilde{B}_x^{od} - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{ev}\right]. \tag{66}$$

Then, as indicated earlier

$$-\omega(\omega-\omega_{di})\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{\xi}_x^{ev} + i\frac{T_e'}{m_i}\frac{k_yB_y}{B^2}\frac{d}{dx}\left[\tilde{B}_x^{od} - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{ev}\right] - \frac{i}{4\pi m_i n}\left[(\mathbf{k}\cdot\mathbf{B})\left(\frac{d^2\tilde{B}_x^{od}}{dx^2}-k_y^2\tilde{B}_x^{od}\right)\right] \simeq -\frac{i}{4\pi m_i n}k_y\frac{d^2B_y}{dx^2}\tilde{B}_x^{ev} \tag{67}$$

and

$$\tilde{B}_x^{od} - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{ev} \simeq \frac{\omega}{\omega-\omega_{*e}}\left(d_I^2+d_e^2\right)\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{B}_x^{od}. \tag{68}$$

We represent the combination of Eqs. (67) and (68) as

$$\Theta_p^{od}(\bar{x},\omega)\tilde{B}_x^{od}(\bar{x}) = -\frac{J_z'}{J_z}\delta_m\tilde{B}_x^{ev}(\bar{x}), \tag{69}$$

where $\Theta_p^{od}(\bar{x},\omega)$ is the relevant odd operator, $J_z' \equiv dJ_z/dx$ and

$$\Theta_p^{od}(\omega=\omega_0)\tilde{B}_x^{od})_0 = 0, \tag{70}$$

where ω_0 and $\tilde{B}_x^{od})_0$ are the eigenvalue and the solution obtained for $\tilde{B}_x^{ev}(\bar{x}) = 0$.

If we consider the realistic case where $\varepsilon_J \equiv -|J_z'\delta_m/J_z| < 1$, we may argue that $\tilde{B}_x^{ev}$ can be of the same order as $\tilde{B}_x^{od})_0$ and in the case Eq. (69), to first order in ε_J becomes

$$\Theta_p^{od}(\omega=\omega_0)\left(\delta\tilde{B}_x^{od}\right) + \left.\frac{\partial\Theta_p}{\partial\omega}\right)_0(\delta\omega)\tilde{B}_x^{od})_0 = \varepsilon_J\tilde{B}_x^{ev}. \tag{71}$$

The odd component of Eq. (15) is

$$-\omega(\omega-\omega_{di})\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{\xi}_x^{od} + \frac{1}{m_i}\frac{k_yB_y}{B^2}\frac{d}{dx}\left[(\mathbf{k}\cdot\mathbf{B})\left(\tilde{T}_{e\|}^{od}-\tilde{T}_{e\perp}^{od}\right)\right] \cong \frac{i}{4\pi m_i n}\left[(\mathbf{k}\cdot\mathbf{B})\left(\frac{d^2\tilde{B}_x^{ev}}{dx^2}-k_y^2\tilde{B}_x^{ev}\right) - k_y\frac{d^2B_y}{dx^2}\tilde{B}_x^{od}\right], \tag{72}$$

where for the odd component of $\tilde{T}_{e\|}$, as given by Eq. (13), the singularity is removed by including the effect of a transverse thermal diffusivity in the relevant thermal energy balance equation. Specifically, we consider the odd component of Eq. (12) given by

$$D_{\perp\|}^e\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{T}_{e\|}^{od} - k_\|^2D_{\|\|}^e\tilde{T}_{e\|}^{od} \simeq -ik_\|D_{\|\|}^eT_e'\frac{\tilde{B}_x^{ev}}{B}. \tag{73}$$

Then, referring to Eq. (10), we have

$$(\mathbf{k}\cdot\mathbf{B})\left(\tilde{T}_{e\|}^{od}-\tilde{T}_{e\perp}^{od}\right) \simeq iT_e'\left[\tilde{B}_x^{ev} - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{od}\right] + \frac{D_{\perp\|}^e(\mathbf{k}\cdot\mathbf{B})}{D_{\|\|}^ek_\|^2}\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{T}_{e\|}^{od}, \tag{74}$$

and Eq. (72) becomes

$$-\omega(\omega-\omega_{di})\left(\frac{d^2}{dx^2}-k_y^2\right)\tilde{\xi}_x^{od} + i\frac{T_e'}{m_i}\frac{k_yB_y}{B^2}\frac{d}{dx} \times \left[\tilde{B}_x^{ev} - i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{od} - i\frac{D_{\perp\|}^e(\mathbf{k}\cdot\mathbf{B})}{D_{\|\|}^ek_\|^2}\left(\frac{d^2}{dx^2}-k_y^2\right)\frac{\tilde{T}_{e\|}^{od}}{T_e'}\right] \cong \frac{i}{4\pi m_i n}\left[(\mathbf{k}\cdot\mathbf{B})\left(\frac{d^2\tilde{B}_x^{ev}}{dx^2}-k_y^2\tilde{B}_x^{ev}\right) - k_y\frac{d^2B_y}{dx^2}\tilde{B}_x^{od}\right]. \tag{75}$$

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For the longitudinal electron momentum conservation equation we refer to Eq. (20) and considering Eq. (68) we have

$$\tilde{B}_x^{ev} \simeq i(\mathbf{k}\cdot\mathbf{B})\tilde{\xi}_x^{od} + \left(d_I^2 + d_e^2\right)\frac{\omega}{\omega-\omega_{*e}}\left(\frac{d^2}{dx^2} - k_y^2\right)\tilde{B}_x^{ev} + i\frac{D_{\perp\parallel}^e}{D_{\parallel\parallel}^e}\frac{\omega_{*T}}{\omega-\omega_{*e}}\frac{\mathbf{k}\cdot\mathbf{B}}{k_\parallel^2 T_e'}\left(\frac{d^2}{dx^2} - k_y^2\right)\tilde{T}_{e\parallel}^{od}, \tag{76}$$

when D_m/ω is neglected in comparison with $(d_I^2 + d_e^2)$. Then, adopting the expression (64) for $\tilde{B}_x^{ev}$ within δ_m we define $\varepsilon_k^2 \equiv k_y^2\delta_m^2$, $d_*^2 \equiv \omega(d_I^2 + d_e^2)/(\omega-\omega_{*e})$, $L_s \equiv B/B_y'$,

$$\alpha_D \equiv \frac{D_{\perp\parallel}^e}{D_{\parallel\parallel}^e}\frac{L_s^2}{k_y^2\delta_m^4}, \tag{77}$$

$$\bar{Y}^{od} \equiv ik_y B_y'\frac{\tilde{\xi}_x^{od}}{\tilde{B}_0^{ev}}\delta_m, \tag{78}$$

and

$$\bar{U}^{od} \equiv -ik_y\delta_m\frac{B_y'}{\tilde{B}_0^{ev}}\frac{\tilde{T}_{e\parallel}^{od}}{T_e'}. \tag{79}$$

Thus, (76) can be rewritten as

$$\frac{d_*^2}{\delta_m^2}\left[\varepsilon_*\frac{d^2\varphi_0}{d\bar{x}^2} - \varepsilon_k^2\right] \cong 1 - \bar{x}\bar{Y}^{od} + \alpha_D\frac{\omega_{*T}}{\omega-\omega_{*e}}\frac{1}{\bar{x}}\left(\frac{d^2}{dx^2} - \varepsilon_k^2\right)\bar{U}^{od}. \tag{80}$$

The asymptotic matching condition with the expression for $\tilde{B}_x$ in the outer (ideal MHD) region is well known as

$$\Delta' = \frac{\varepsilon_*}{\delta_m}\int\frac{d^2\varphi_0}{d\bar{x}^2}d\bar{x}, \tag{81}$$

where $\Delta' \sim 1/a_z \equiv -J_z'/J_z$ represents the jump of the logarithmic derivative of $\tilde{B}_x^{ev}$ for $x - x_0 \to \pm 0$ in the outer region. This implies

$$\varepsilon_* \sim \delta_m\Delta', \tag{82}$$

and Eq. (73) for $\tilde{T}_{e\parallel}^{od}$ becomes

$$\alpha_D\left(\frac{d^2}{dx^2} - \varepsilon_k^2\right)\bar{U}^{od} - \bar{x}^2\bar{U}^{od} \simeq -\bar{x}. \tag{83}$$

The solution of Eq. (83), for $\varepsilon_k^2 \ll 1$, is well-known.[2] We can then rewrite Eq. (75) as

$$\Gamma_0^2\left(\frac{d^2}{d\bar{x}^2} - \varepsilon_k^2\right)\bar{Y}^{od} - \alpha_A^0\frac{d}{d\bar{x}}\left[\bar{x}\left(\bar{Y}^{od} - \bar{U}^{od}\right)\right] \simeq \bar{x}\left[\bar{x}\bar{Y}^{od} - 1 + \frac{\omega_{*T}}{\omega-\omega_{*e}}\left(1 - \bar{x}\bar{U}^{od}\right)\right] - \frac{d_*^2}{a_z\delta_m}\frac{\tilde{B}_0^{od}}{\tilde{B}_0^{ev}}\bar{x}W_0(\bar{x}^2), \tag{84}$$

where $\alpha_A^0 \equiv -T_e'B_y/(T_eB_y')$,

$$\Gamma_0^2 \equiv -\frac{\omega(\omega-\omega_{di})\alpha_A^0}{k_y^2\delta_m^2\omega_H^2\varepsilon_A}, \tag{85}$$

and

$$\delta_m^2 = \varepsilon_A\frac{d_*^2}{\alpha_A^0}. \tag{86}$$

We have used Eq. (83) in the above derivations.

The condition for the asymptotic matching of the solution for $\tilde{B}_x^{ev}$ inside the δ_m-region with that (ideal MHD) outside the δ_m-region involves the integral

$$\int_{-\infty}^{+\infty}d\bar{x}\left[1 - \bar{x}\bar{Y}^{od} - \frac{\omega_{*T}}{\omega-\omega_{*e}}\left(1 - \bar{x}\bar{U}^{od}\right)\right] \equiv \Im. \tag{87}$$

Then, referring to Eqs. (80) and (87), we have

$$\Delta' = \frac{\delta_m}{d_*^2}\Im. \tag{88}$$

Since δ_m can be taken as the width of localization of $\tilde{B}_x^{od}$, we consider

$$\delta_m \gg d_*^2\Delta' \sim \frac{d_*^2}{a_z}.$$

Then, referring to Eq. (82), we may consider the condition

$$\int d\bar{x}\left[1 - \bar{x}\bar{Y}^{od} + \frac{\omega_{*T}}{\omega-\omega_{*e}}\frac{\alpha_D}{\bar{x}}\frac{d^2U^{od}}{d\bar{x}^2}\right] \simeq 0. \tag{89}$$

Clearly, the composite solution proposed in this section has the advantage of including the effects of the local current density gradient, of extending well beyond the reconnection layer, and of involving a size of this layer that can be significant even when large macroscopic distances are involved such as in the case of space physics.

The process resulting in the formation of magnetic islands due to the combination of symmetric and antisymmetric $\tilde{B}_x(x-x_0)$ reconnection may be referred to as asymmetric reconnection. Finally, we note that the theory of the "internal (reconnecting) kink mode"[2] involves a composite solution made of the sum of $\tilde{\xi}_r(r-r_0)$ even and $\tilde{\xi}_r(r-r_0)$ odd. We note that the drift tearing mode has also a composite solution but the $\tilde{B}_x$-odd component of it does not contribute to the relevant growth rate.

XII. CONCLUSIONS

A major finding of the present paper is that magnetic field fluctuations can be generated in the presence of a significant electron temperature gradient, a reconnection process that converts electron thermal energy into magnetic field energy. We note that the standard theoretical model for the origin of the cosmic magnetic fields is the generation of weak seed magnetic fields due to the "Biermann Battery" effect, and then the amplification of the seed fields via turbulent dynamo mechanism to the level consistent with the current observations.[10] Since there are plasmas in the Universe with considerable electron thermal energy contents, we suggest that the reconnection process described in the present

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paper may also play a significant role in the origin of the observed strong cosmic magnetic fields.

Referring to the Earth's magnetosphere there is real evidence of collisonless magnetic reconnection processes in the presence of electron temperature gradients and anisotropies.[11] Clearly the theory given here needs to be extended to field geometries that are relevant to field configurations that may represent those resulting from *in situ* measurements in the Earth's magnetotail[12] or to the current sheet present in the Heliosphere. In fact, the analysis described in the present paper has been extended to the case where a neutral sheet configuration with temperature gradients is considered,[13] from which reconnecting modes can emerge. We observe that in Ref. 14 a magnetic field geometry that should be close to those inferred from the observations in the Earth's magnetotail is adopted.

The best agreement between theory and experiments concerning the onset of magnetic reconnection in laboratory plasmas is (probably) represented by the theory of the internal kink mode.[2] A proposed explanation of the observed accelerated reconnection rate following the onset[15] is that, in this case, reconnection involves the formation of a relatively large magnetic island and a local steepening of the electron temperature gradient. Then, the effects of the endogenous reconnecting mode associated with electron temperature gradient are compounded with those (e.g., total plasma pressure gradient) that lead to the onset of reconnection.

Endogenous modes associated with a finite electron temperature gradient may be sustained by the electron temperature heating rate due to the charged reaction products in a fusion burning plasma.[16] In this case, the longitudinal thermal conductivity on selected rational magnetic surfaces,[16] referring to a toroidal confinement configurations, is decreased, relative to its collisional value, by the effects of reconnection.

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